CS341 · COMPUTER GRAPHICS

# Midterm 461 Practice

20 marks

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## 01 Questions

Q1 SHORT ANSWER

1. Define computer graphics.
2. What is the difference between absolute and relative coordinate input?
3. A screen has a horizontal length of 10 inches and a vertical height of 5 inches. What is the aspect ratio given that the screen resolution is 1920 × 1080 points?

## Q2 TRUE / FALSE

1. Plasma displays are more energy-efficient than LCD displays. (  )
2. M + Y + C = White. (  )
3. A raster scan device scans the screen from vertical to horizontal. (  )
4. Raster scan is more efficient than random scan in drawing lines and shapes. (  )
5. The refresh rate is the measure of how many times the image on a screen is updated per second. (  )
6. Graphic applications are used only for entertainment. (  )

## Q3 CHOOSE / COMPLETE

1. The standard design of a video monitor is the _____.
2. Subtractive mixing is the principle of the _____ color model.
3. The frequency at which a picture is redrawn on the screen is referred to as the _____.
4. A goal of computer graphics: _____.
5. The rays emitted by a cathode ray tube are _____.
6. Applications of computer graphics: CAD / Medical visualization / Entertainment / _____.
7. Digital displays use the _____ color model.

**Q4** FILL IN THE BLANKS

1. Yellow + Magenta = _____
2. Magenta + Cyan = _____
3. Green + Blue = _____
4. _____ is the maximum number of points/pixels that can be displayed (without overlap) on a CRT.
5. _____ is defined as the time that it takes the emitted light from the screen to decay to one-tenth of its original intensity.
6. _____ is the set of all colors that can be produced from mixing the primary colors.
7. _____ is white pigment added to pure pigment.
8. _____ is the consequence of adding both white and black pigments to pure pigments.

**Q5** LIST

1. List four hard-copy devices.
2. List four applications of computer graphics.
3. RGB stands for:

**Q6** BRESENHAM

Using Bresenham's algorithm, find the pixels of the line starting at (3, 3) and ending at (10, 7).

| k | $p_k$ | Decision | Next pixel | $p_{k+1}$ |
|---|---|---|---|---|
| 0 | | | | |
| 1 | | | | |
| 2 | | | | |
| 3 | | | | |
| 4 | | | | |
| 5 | | | | |
| 6 | | | | |

